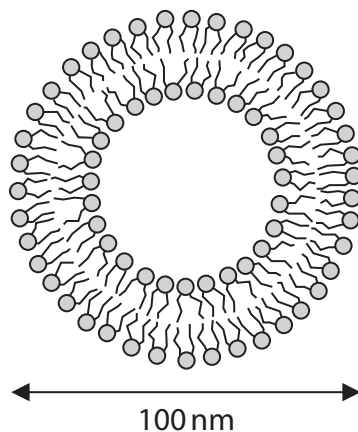


- 5 Liposomes are spherical structures composed of phospholipids. They can be made by adding phospholipids to water.

Liposomes can be used to study membrane permeability.

- (a) The diagram shows a liposome.



- (i) Calculate the volume of this liposome, using the formula

$$V = \frac{4}{3}\pi r^3$$

(2)

Answer

- (ii) Explain the arrangement of phospholipids in liposomes.

(2)

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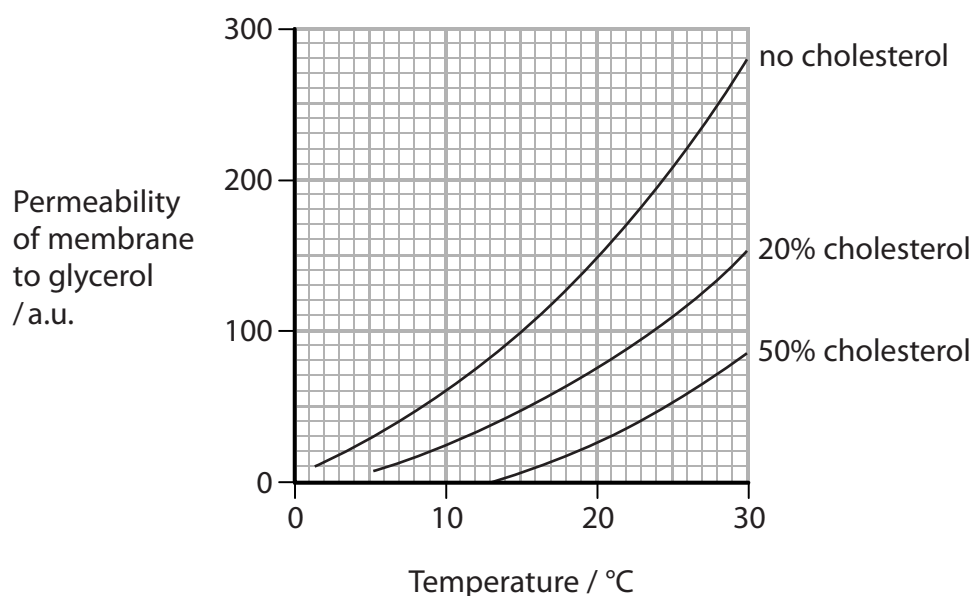
(b) The presence of cholesterol in the membrane affects membrane permeability.

A student investigated the effect of cholesterol on the permeability of liposomes to glycerol at different temperatures.

Liposomes were made by replacing 20% and 50% of the phospholipid with cholesterol.

Liposomes without cholesterol were also made.

The graph shows the results of this investigation.



(i) How does glycerol pass through the liposome membrane?

(1)

- ☒ **A** active transport
- ☒ **B** diffusion
- ☒ **C** endocytosis
- ☒ **D** osmosis



(ii) Describe the effects of cholesterol and temperature on membrane permeability, as shown in the graph.

(3)

(iii) Explain why cholesterol and temperature affect membrane permeability.

(2)

(Total for Question 5 = 10 marks)

